

Linux Privilege Escalation - Docker & Kubernetes Attacks Cheatsheet

Privilege escalation attacks in containerized environments like Docker and Kubernetes are serious threats to system security. Both platforms, while designed to isolate and secure applications, can be vulnerable to attacks that allow an attacker to break out of containers or escalate privileges. Understanding these risks and how to exploit them is crucial for security professionals and system administrators to harden their environments. This cheatsheet will guide you through common Docker and Kubernetes privilege escalation techniques, providing resources, attack vectors, and security tools to enhance your awareness and defense strategies.

Docker Privilege Escalation Attacks

Attack	Description	Resources
Docker Breakout	Escaping the container and gaining access to the underlying host system.	HackTricks - Docker Breakout
Kernel Vulnerabilities	Exploiting kernel vulnerabilities to gain elevated privileges on the host machine from within Docker containers.	Trail of Bits - Understanding Docker Container Escapes
Improper Container Configurations	Misconfigured Docker containers or permissions that allow an attacker to access privileged resources or escalate privileges.	tbhaxor - Container Breakout (Part 1)

Helpful Video	Description
Docker Escapes - Live Demo	A detailed live demo of Docker escapes.

Kubernetes Privilege Escalation Attacks

Attack	Description	Resources
Pod/Container Escape via Privileged Pod	Escaping a privileged pod to gain access to the host system.	See the command below for a sample attack.
Exploiting Insecure RBAC	Weak Role-Based Access Control (RBAC) configurations allowing unauthorized users to escalate privileges.	Kubernetes Goat - A vulnerable-by-design Kubernetes cluster for practice.
Kubernetes Cluster Vulnerabilities	Leveraging known vulnerabilities in Kubernetes components or misconfigurations to escalate privileges.	Kube-Hunter by Aqua Security - Tool for hunting Kubernetes cluster vulnerabilities.

Helpful Video	Description
Kubernetes Security	Video walkthrough of Kubernetes security practices.

POD/CONTAINER ESCAPE VIA PRIVILEGED POD ATTACK EXAMPLE:

Command
<pre>kubectl run r00t --restart=Never -ti --rm --image lol --overrides '{"spec":{"hostPID": true, "containers":[{"name":"1","image":"alpine","command":["nsenter","--mount=/proc/1/ns/mnt","--","/bin/bash"],"stdin":true,"tty":true,"securityContext":{"privileged":true}}]}'</pre>

Security Tools for Mitigation

Tool	Description	Link
Kube-Hunter	A tool to hunt for vulnerabilities in Kubernetes clusters.	Kube-Hunter by Aqua Security
Falco	A cloud-native runtime security tool for Kubernetes and Docker.	Falco Security
Docker Bench for Security	A script that checks for common Docker security issues and provides recommendations.	Docker Bench for Security

General Mitigation and Hardening Tips

Mitigation	Description
Use User Namespaces	Isolate containers by mapping user IDs in the container to different IDs on the host.
Limit Privileges	Avoid running containers as root, and enforce least privilege by limiting container capabilities.
Security Contexts in Kubernetes	Implement Kubernetes security contexts to restrict pod privileges and isolate workloads.
Use SELinux or AppArmor	Enable SELinux or AppArmor to enforce security policies that prevent container escapes.
Network Policies	Define strict network policies to limit inter-container communication.